Zilu Zheng

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EDUCATION

University of California - San Diego **Sep. 2026 - Expected Jul. 2029**
Bachelor of Science, Aerospace Engineering *La Jolla, CA*
Relevant Coursework (In Progress): Thermodynamics, Fluid Mechanics, Material Science
Relevant Coursework (Planned, Winter & Spring 2027): Aerodynamics, Solid Mechanics, Signals and Systems

De Anza College **Jul. 2024 - Jul. 2026**
GPA: 3.9/4.0; Relevant Coursework (Completed): Circuit Analysis & Lab, Statics *Cupertino, CA*

SKILLS

Technical Skills: Creo Parametric (Top-down Design, Surfacing, Simulate/FEA, Sheetmetal), GD&T, FDM 3D Printing Programming & Tools: Python, HTML/CSS, Raspberry Pi, YOLO object detection

PROJECTS & EXPERIENCE

UCSD Engineering Teams (Current) **Oct. 2026 – Present**
NASA RASC-AL Team (AIAA) | Baja SAE Chassis Team *La Jolla, CA*

- Selected via application to a 10-person team developing a concept proposal for NASA's 2027 RASC-AL Competition, a student design challenge focused on lunar surface technologies
- Assigned to the chassis subteam of UCSD's Baja SAE team, responsible for vehicle frame design for the SAE Baja collegiate competition

Libreye - Foothill Innovation Challenge **May 2026 – Jun. 2026**
Design Engineer *Los Altos Hills, CA*

- Owned structural design, manufacturing, and assembly of a compact, modular enclosure for a cloud-free security camera prototype on a 3-person team spanning software, hardware, and mechanical
- Modeled the housing in Creo Parametric (top-down design, surfacing, parametric relations) and iterated prototype variants through FDM 3D printing in PLA
- Increased shell stiffness 67% in Creo Simulate FEA (max deflection 0.97 → 0.58 mm under 250 N side load, fixed mating face) for only +0.8 g (+2.9%) mass by adding ribs and fillets; peak von Mises stress fell 12% (26 → 23 MPa)

Two-Stage Rocket - De Anza College **Jan. 2026 – May 2026**
Structures Lead *Cupertino, CA*

- Led a 10-person structures subteam building a two-stage high-power rocket (dual L1 motors), personally fabricating flight hardware and training members in Creo CAD
- Designed and 3D-printed (FDM) the avionics bay, nosecone, and structural mounts in Creo Parametric to a 15 g acceleration requirement from OpenRocket simulations
- Redesigned bay packaging and mounts as electronics, wiring, and part specs changed, coordinating with the avionics team to keep the structure compatible with the flight computer

AI Smart-fan - Infineon Technologies **Jan. 2025 - May 2025**
Engineering Intern *San Jose, CA*

- Built a person-tracking, gesture-controlled smart fan on a 4-person team, integrating Infineon PSoC 6 gesture control with YOLOv10n person tracking on a Raspberry Pi 4B
- Wrote the initial Python person-detection prototype (~0.5 fps on the Raspberry Pi), which teammates optimized to ~10 fps for real-time fan tracking
- Designed the mobile-friendly web app front end (HTML/CSS) for switching between tracking, gesture, and manual modes
- Won "Best Use of DEEPCRAFT™ Ready-Models" ($500) as a team in an Infineon hackathon with 460 participants, after a live demo to Infineon engineers
- Co-authored Hackster.io documentation (500+ views) covering system architecture and build instructions, plus a video demo Link to the project